MODULE 3 BLOOD VESSELS

Blood Vessels, Structure and Types.

BLOCK 3 · MODULE 4: VESSEL ANATOMY AND HISTOLOGY

A reference for the blood vessels video and lab. This chapter covers the three tunics of a vessel wall, the five vessel types, the kinds of artery, capillary, and vein, and the circulatory routes blood follows. Read it with a slide in front of you; almost all of it is visible histology.

BY THE END

1. Name the three tunics of a vessel wall and what each contains.
2. Compare the five vessel types from artery to vein.
3. Distinguish elastic, muscular, and arteriolar arteries on a slide.
4. Distinguish the three kinds of capillary and say where each is found.
5. Describe the circulatory routes, including portal systems and anastomoses.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Blood vessels

drsrennie-stack.github.io/new-build-bio4-solano/blood-vessels-concept-videos.html

Blood Vessels, an Overview

Blood vessels are the body's plumbing. Blood leaves the heart in arteries, reaches the tissues through capillaries, and returns in veins.

THE BASIC VOCABULARY

Structure	What it is
Blood vessel	A tube that carries blood through the body
Artery	A vessel that carries blood away from the heart
Vein	A vessel that carries blood toward the heart
Capillary	The smallest vessel, where exchange between the blood and the tissues takes place
Lumen	The hollow interior of a vessel, the channel the blood flows through
The vascular sequence	Arteries branch into arterioles, then capillaries, which merge into venules, then veins

The Three Tunics

The wall of an artery or a vein is built in three concentric layers, the tunics. A capillary wall is just the innermost layer.

COMPARE THE THREE

Tunic	Position	Tissue	Note
Tunica intima	The innermost layer	A lining of simple squamous endothelium on a basement membrane	The only layer a capillary has. Continuous with the endocardium of the heart
Tunica media	The middle layer	Smooth muscle and elastic fibers	The thickest layer in arteries. Its smooth muscle sets the vessel's diameter
Tunica externa	The outermost layer	Mostly collagen fibers, with nerves and small vessels	Anchors the vessel to the surrounding tissue. The thickest layer in veins

THREE MORE STRUCTURES IN THE WALL

Structure	What it is
Internal elastic lamina	A fenestrated sheet of elastic fibers at the boundary between the tunica intima and the tunica media
External elastic lamina	The elastic sheet separating the tunica media from the tunica externa
Vasa vasorum	The small vessels within the tunica externa that supply the wall of a large vessel. Easily seen on the aorta

KEY POINT

The two elastic laminae are the boundaries of the tunica media. If you can find them on a slide you have found the middle layer, and the thickness of what lies between them tells you whether you are looking at an artery or a vein.

The Vessel Types

Blood passes through five kinds of vessel in series, each with a wall built for its job.

COMPARE THEM

Vessel	Direction of flow	Wall	Role
Arteries	Away from the heart	Thick walls with a heavy tunica media	Carry blood under high pressure
Arterioles	Away from the heart, the smallest arteries	Thinner walls, one to two layers of smooth muscle	Adjust how much blood enters a capillary bed. The chief site of resistance
Capillaries	Connect arterioles to venules	A single layer, the tunica intima only	The exchange vessels; gases and nutrients cross here
Venules	Toward the heart, the smallest veins	Thin, porous walls	Collect blood drained from the capillaries; the site where white blood cells leave the bloodstream
Veins	Toward the heart	Thinner walls and a wide lumen, with valves	Carry blood under low pressure

Arteries

Arteries carry blood away from the heart under high pressure. They come in three sizes, and the smallest control how much blood reaches the tissues.

THE THREE SIZES OF ARTERY

Artery	Tunica media	Examples	Function
Elastic artery	Thick, dominated by elastic lamellae; well-defined internal and external elastic laminae	Aorta, pulmonary trunk, brachiocephalic, subclavian, common carotid, common iliac	Stretches with each heartbeat and recoils between beats, propelling blood while the ventricles relax
Muscular artery	Mostly smooth muscle, up to forty layers; thin external elastic lamina	Femoral, axillary, brachial, radial	Distributes blood to specific organs and maintains vascular tone
Arteriole	One or two layers of smooth muscle; internal elastic lamina disappears at the terminal end	Microscopic, roughly four hundred million of them	Regulates blood flow into a capillary bed and sets resistance

HOW AN ARTERIOLE CONTROLS FLOW

Structure	What it is
Vasoconstriction	The narrowing of a vessel as the smooth muscle of its wall contracts
Vasodilation	The widening of a vessel as the smooth muscle of its wall relaxes
Metarteriole	The tapered terminal end of an arteriole, at the junction with a capillary bed
Precapillary sphincter	A ring of smooth muscle at the metarteriole that opens or closes the entrance to a capillary
Vascular tone	The partial contraction a vessel holds at rest, which stiffens the wall and maintains pressure

Capillaries

Capillaries are the only vessels thin enough for exchange. They lack a tunica media and a tunica externa entirely: the wall is endothelium and a basement membrane. There are three types, leakier as the tissue's job demands.

THE THREE CAPILLARY TYPES

Capillary type	Wall	Where found
Continuous capillary	An unbroken endothelial lining, interrupted only by intercellular clefts	The most common type. Muscle, skin, lungs, and the central nervous system
Fenestrated capillary	An endothelium with small pores, called fenestrations	Where rapid absorption or filtration happens: kidney, small intestine, choroid plexus, most endocrine glands
Sinusoid	A wide, winding vessel with large fenestrations, an incomplete basement membrane, and wide clefts	Where even whole cells must cross: liver, spleen, red bone marrow, anterior pituitary, adrenal glands

HOW CAPILLARIES ARE ORGANIZED

Structure	What it is
Capillary bed	A network of ten to one hundred capillaries supplied by a single metarteriole
Microcirculation	The flow from metarteriole to capillaries to postcapillary venule
Thoroughfare channel	The distal end of a metarteriole, which has no smooth muscle and acts as a direct route from arteriole to venule
Vascular shunt	A direct channel that lets blood bypass the capillaries of a bed when exchange is not needed
Avascular tissues	Tissues with no capillaries at all: all covering and lining epithelia, the cornea and lens, and cartilage

KEY POINT

Capillary type follows the job of the tissue. Tight where things must be kept out, windowed where fluid must move fast, and frankly leaky where whole cells have to pass. One vessel name, three builds, three jobs.

Veins

Veins return blood to the heart under low pressure. Because the pressure is low, they rely on valves and on nearby muscles to keep blood moving the right way.

THE VENOUS SIDE

Structure	What it is
Venule	The smallest vein, collecting blood as it leaves a capillary bed
Postcapillary venule	The venule that first receives blood from the capillaries. Loosely joined and very porous
Muscular venule	The next vessel along, with one or two layers of smooth muscle
Vein	A vessel returning blood to the heart, with a thinner wall and a wider lumen than the matching artery
Venous valves	Flaps of the tunica intima, common in the limb veins, that keep blood from flowing backward
Skeletal muscle pump	Contracting limb muscles that squeeze the deep veins and push blood toward the heart
Blood reservoirs	The veins as a group, which at rest hold the largest share of the body's blood, roughly sixty percent
Vascular sinus	A vein with a thin endothelial wall and no smooth muscle, supported by surrounding dense connective tissue. The dural venous sinuses and the coronary sinus

KEY POINT

A vein has the same three tunics as an artery, in different proportions. Thin media, thick externa, wide lumen. On a slide the artery holds its round shape and the vein looks collapsed, and that is usually the fastest way to tell the pair apart.

Circulatory Routes

Most blood follows the simple loop of artery to capillary to vein, but a few routes are arranged differently.

THE ROUTES WORTH NAMING

Structure	What it is
Pulmonary circuit	The route from the right side of the heart to the lungs and back to the left side
Systemic circuit	The route from the left side of the heart out to the body and back to the right side
Portal system	An arrangement in which blood passes through two capillary beds in series before returning to the heart
Hepatic portal circulation	The portal system that routes blood from the digestive organs through the liver
Hypophyseal portal system	The portal system linking the hypothalamus to the anterior pituitary
Anastomosis	A place where two vessels join, giving blood an alternate route if one path is blocked
Arteriovenous anastomosis	A direct connection from an artery to a vein that bypasses the capillaries

WHY THE WALL BUILT FOR THE JOB MATTERS

Blood is drawn and IV lines are placed in **veins**, not arteries. A vein sits closer to the surface, has a thinner wall and a wider lumen, and runs at low pressure, so it is easy to enter and slow to bleed. Its **valves** can even be seen as small bumps along a forearm vein.

The **capillary** type is matched to the tissue. The leaky **sinusoids** of the liver and spleen let whole blood cells pass, while the tight **continuous capillaries** of the brain keep most substances out. The same vessel name, built three different ways, does three different jobs.

The elastic recoil of the **aorta** keeps blood moving between heartbeats. When elastic arteries stiffen with age, that cushioning is lost, and the strain shows up downstream in the smaller vessels.